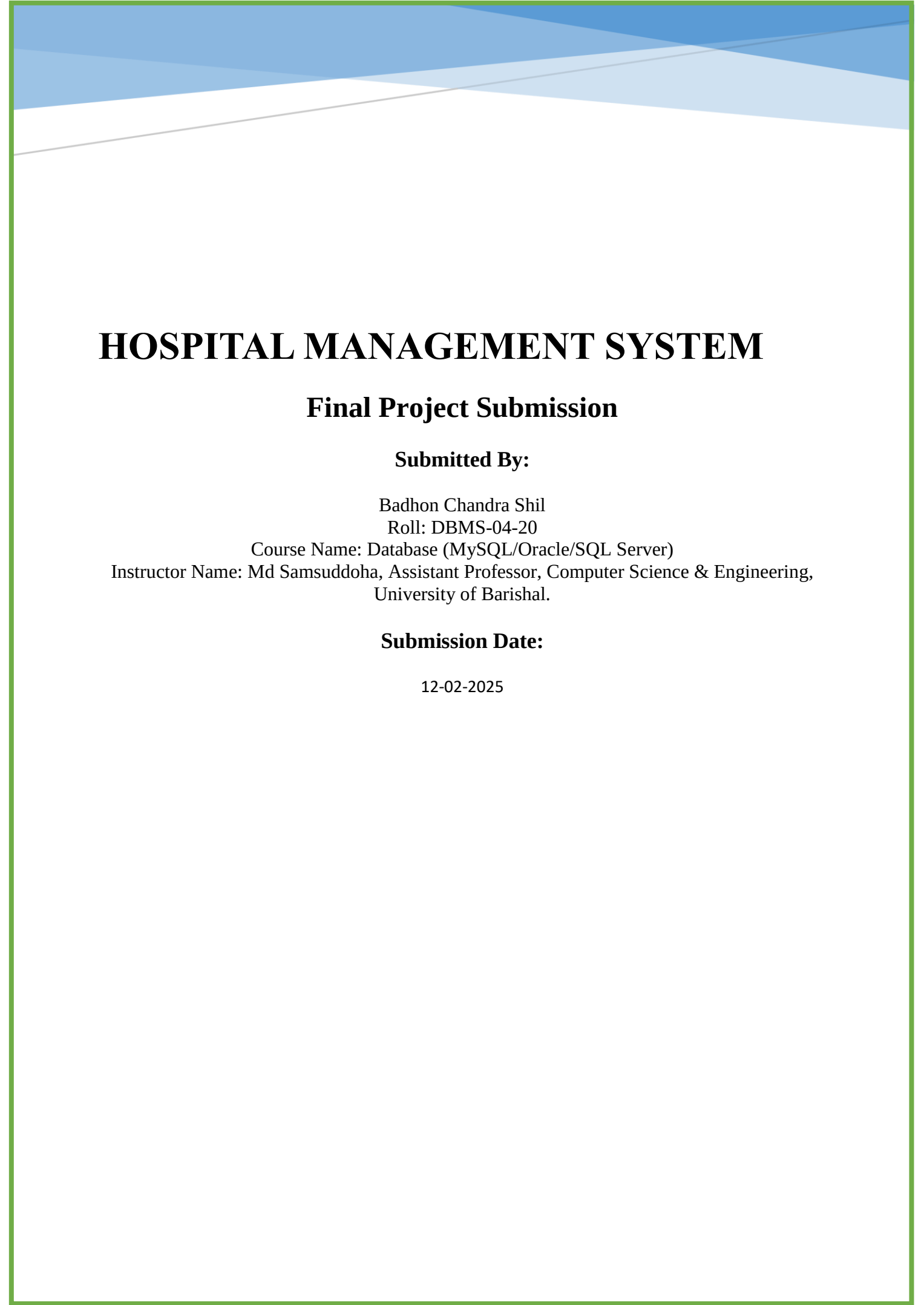




HOSPITAL MANAGEMENT SYSTEM

Final Project Submission

Submitted By:

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Roll: DBMS-04-20

Course Name: Database (MySQL/Oracle/SQL Server)

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Hospital Management System

Project (ERD Design)

Project (HMS):

I have to develop a Hospital Management System database for a hospital. The system will manage **patients**, **doctors**, **appointments**, **medical records**, and **billing**. The hospital has multiple doctors specializing in different fields, and each doctor can have multiple appointments with other patients. Patients can have various medical records and billing details.

Steps of Constructing ERD

1. Identify the Entities Required
 2. Identify the Attributes and Primary Key for each Entity
 3. Identify the Relationship Needed
 4. Identify the Cardinality Ratio and Participation
 5. Draw the Diagram
-

Step 1: Identify Entities

I have to develop a database for a hospital including some information on **Patients**, **Doctors**, **Appointments**, **Medical Records**, and **Billing**.

- Each **patient** can book an **appointment** with a **doctor**.
 - **Doctors** can have multiple **appointments** but each appointment is assigned to a single **patient**.
 - Each **patient** has a unique medical history stored in **medical records**.
 - **Billing** keeps track of charges for hospital services.
-

Step 2: Identify Attributes and Primary Key

Entities & Attributes

- **Patients** (*PatientID*, Name, Age, Gender, Address, Phone)
- **Doctors** (*DoctorID*, Name, Specialization, Phone)
- **Appointments** (*AppointmentID*, PatientID, DoctorID, AppointmentDate, Status)
- **Medical Records** (*RecordID*, PatientID, Diagnosis, Treatment, DateRecorded)

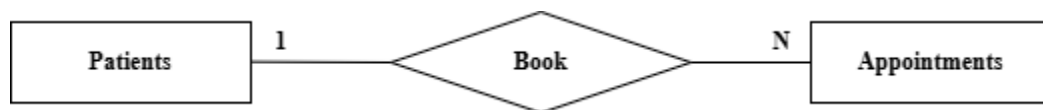
- **Billing** (*BillID*, PatientID, Amount, PaymentStatus)
-

Step 3: Identify the Relationship Needed

1. **Patients book Appointments**
 2. **Doctors manage Appointments**
 3. **Patients have Medical Records**
 4. **Patients have Billing records**
-

Step 4: Identify the Cardinality Ratio and Participation

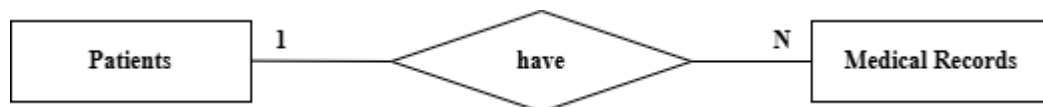
1. **Patients to Appointments** → One-to-Many (Total for Appointments)



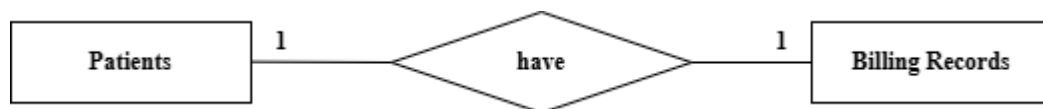
2. **Doctors to Appointments** → One-to-Many (Total for Appointments)



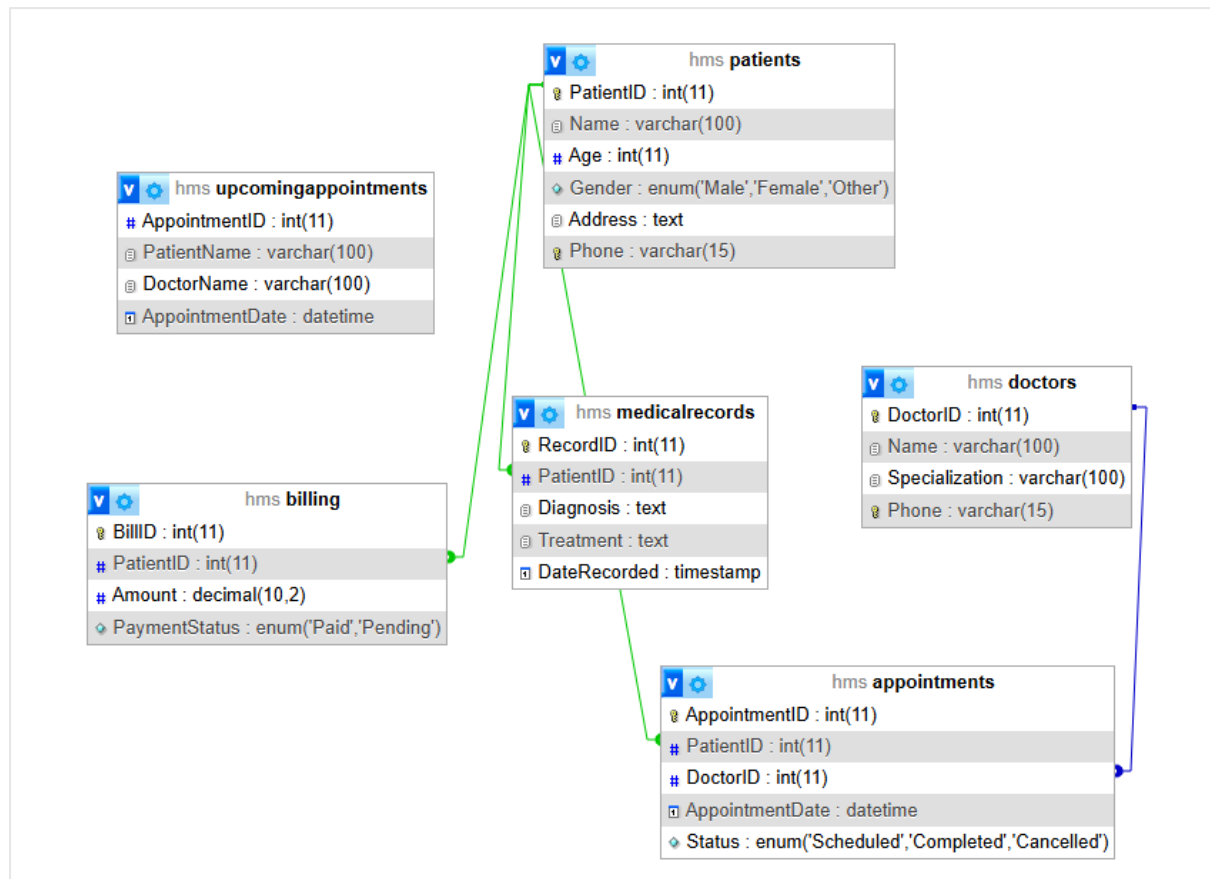
3. **Patients to Medical Records** → One-to-Many (Total for Medical Records)



4. **Patients to Billing** → One-to-One (Total for Billing)



Step 5: Implementing the Schema Diagram



Database Schema (Tables & Relationships)

Patients Table

```
CREATE TABLE Patients (  
    PatientID INT PRIMARY KEY AUTO_INCREMENT,  
    Name VARCHAR(100) NOT NULL,  
    Age INT CHECK (Age > 0),  
    Gender ENUM('Male', 'Female', 'Other') NOT NULL,  
    Address TEXT,  
    Phone VARCHAR(15) UNIQUE NOT NULL  
);
```

Doctors Table

```
CREATE TABLE Doctors (
    DoctorID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    Specialization VARCHAR(100) NOT NULL,
    Phone VARCHAR(15) UNIQUE NOT NULL
);
```

Appointments Table

```
CREATE TABLE Appointments (
    AppointmentID INT PRIMARY KEY AUTO_INCREMENT,
    PatientID INT,
    DoctorID INT,
    AppointmentDate DATETIME NOT NULL,
    Status ENUM('Scheduled', 'Completed', 'Cancelled') DEFAULT 'Scheduled',
    FOREIGN KEY (PatientID) REFERENCES Patients(PatientID) ON DELETE CASCADE,
    FOREIGN KEY (DoctorID) REFERENCES Doctors(DoctorID) ON DELETE CASCADE
);
```

Medical Records Table

```
CREATE TABLE MedicalRecords (
    RecordID INT PRIMARY KEY AUTO_INCREMENT,
    PatientID INT,
    Diagnosis TEXT NOT NULL,
    Treatment TEXT NOT NULL,
    DateRecorded TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (PatientID) REFERENCES Patients(PatientID) ON DELETE CASCADE
);
```

Billing Table

```
CREATE TABLE Billing (
    BillID INT PRIMARY KEY AUTO_INCREMENT,
    PatientID INT,
    Amount DECIMAL(10,2) NOT NULL,
    PaymentStatus ENUM('Paid', 'Pending') DEFAULT 'Pending',
    FOREIGN KEY (PatientID) REFERENCES Patients(PatientID) ON DELETE CASCADE
);
```

Stored Procedures, Triggers, and Views

Stored Procedure: Add New Appointment

```
DELIMITER $$
CREATE PROCEDURE AddAppointment(IN p_PatientID INT, IN p_DoctorID INT, IN
p_Date DATETIME)
BEGIN
    INSERT INTO Appointments (PatientID, DoctorID, AppointmentDate)
    VALUES (p_PatientID, p_DoctorID, p_Date);
END $$
DELIMITER ;
```

Trigger: Update Payment Status on Billing

```
DELIMITER $$
CREATE TRIGGER After_Bill_Paid
AFTER UPDATE ON Billing
FOR EACH ROW
BEGIN
    IF NEW.PaymentStatus = 'Paid' THEN
        UPDATE Appointments SET Status = 'Completed' WHERE PatientID =
NEW.PatientID;
    END IF;
END $$
DELIMITER ;
```

View: Upcoming Appointments

```
CREATE VIEW UpcomingAppointments AS
SELECT a.AppointmentID, p.Name AS PatientName, d.Name AS DoctorName,
a.AppointmentDate
FROM Appointments a
JOIN Patients p ON a.PatientID = p.PatientID
JOIN Doctors d ON a.DoctorID = d.DoctorID
WHERE a.AppointmentDate > NOW();
```

Conclusion

This project provides a structured Hospital Management System using MySQL. It includes essential database functionalities, constraints, stored procedures, triggers, and views for efficient management.
